KENDRIYA VIDYALAYA SANGATHAN, BENGALURU REGION

केन्द्रीय विद्यालय संगठन, बेंगलुरु संभाग

FIRST PRE BOARD EXAMINATION 2025-26(Marking scheme)

प्रथम प्री बोर्ड परीक्षा 2025-26

CLASS:X MAXIMUM MARKS:50

Subject:Artificial Intelligence Time: 2Hrs

	SECTION A	
1	**Answer any 4 out of 6**	4X1=4
I	____________is a type of communication which is conveyed only through images or pictures : VISUAL COMMUNICATION	
Ii	B. Relaxation techniques and positive thinking	
Iii	B. Strength – Communication; Weakness – Time management	
Iv	Answer: (b) To detect and remove malicious programs	
V	B. Arjun is in self-employment, Neha is in wage employment.	
Vi	C. Adopting renewable energy and recycling resources	
2	**Answer any 5out of 6**	5X1=5
I	Answer: B. Data Acquisition	
Ii	Answer: B. Learning-based model	
Iii	Answer: (b) Both A and R are true and R is not the correct explanation for A	
Iv	Answer: (a) Both A and R are true and R is the correct explanation for A	
V	B. Computer Vision	
vi	Answer: A. Translating text from one language to another and voice assistants understanding commands	
3	**Answer any 5out of 6**	5X1=5
I	Answer: C. Fairness	
Ii	Answer: C. Training	
Iii	True Positives(TP): 90 False Positives(FP): 50. True Negatives(TN): 820 False Negatives (FN): 80	
Iv	**Confusion Matrix for Fraud Detection System**	

Confusion Matrix	Reality: Yes (Fraudulent)	Reality: No (Legitimate)
Prediction: Yes (Fraudulent)	**TP = 80**	**FP = 30**
Prediction: No (Legitimate)	**FN = 40**	**TN = 850**

V	Answer: B. Recognizing and interpreting visual data from camera feeds	
vi	Answer: A. Sentiment analysis	
4	**Answer any 5out of 6**	5X1=5
I	Answer: C. Model Evaluation	
Ii	Answer: B. Overfitting	
Iii	C) Predicts a continuous value as output	
Iv	Answer: A. Reduces manual monitoring and increases efficiency	

V	Answer: B. Speech-to-text and natural language understanding	
vi	Answer: C. Machine translation	
5	**Answer any 5out of 6**	5X1=5
I	Answer: B. Transparency	
Ii	Answer: B. Modify the goal to ensure privacy protection	
Iii	Answer: B. Testing on a separate dataset	
Iv	Answer: A. Supervised learning	
V	(b) Precision & Recall	
vi	Answer: A. Text classification	
	SECTION B	
	Answer any 3 out of given 5 question on employability skills(answer each question in20-30 words)	3X2=6
6	a) Green skill demonstrated: Waste management / Recycling and reuse. b) Explanation: These practices reduce environmental pollution, conserve resources, and promote sustainable development by encouraging eco-friendly production.	
7	Self-confidence – She believes in her abilities and ideas. Risk-taking ability – She is willing to take calculated risks to start her business. Or any other relevant quality .	
8	a) Self-management skills: Time management and self-motivation. b) Explanation: These skills help Amit stay organized, focused, and productive, reducing stress and improving his overall performance.	
9	Ict Practical steps to maintain computers: 1. Regular software updates and antivirus scans: Keep the operating system and antivirus updated to protect against security threats. 2. Clean hardware components: Gently clean the keyboard, monitor, and CPU to prevent dust accumulation that can cause overheating. 3. Organize and back up data: Delete unnecessary files, empty the recycle bin, and back up important data regularly to avoid data loss. (Optional additional point: Avoid installing unnecessary software or using external devices without scanning for viruses.)	
10	Pause and reset: Briefly pause to allow yourself and the audience to reset. This gives you time to assess the situation and gives the audience a moment to process what was just said ● Rephrase and simplify: Restate the last point using different, clearer words. You can use a simpler example or an analogy to explain it. Change your tone: Experiment with your tone of voice. You can try speaking more slowly, or slightly raising your volume to emphasize a key point, as your tone conveys a lot of information Engage with the audience Make eye contact with different people around the room, including those who looked confused. Use your body language to appear approachable and relaxed, which can encourage others to ask questions	

	Answer any 4 out of given 6 question on employability skills(answer each question in20-30 words)	4X2=8
11	a) Bioethics: Bioethics refers to the study of ethical issues and moral principles related to biology, medicine, and AI applications that affect human life, such as privacy, safety, fairness, and consent. b) Example: Ensuring patient data privacy while developing an AI system for medical diagnosis.	
12	Expected Answer: a) Rule-based AI: The system that follows fixed rules. Learning-based AI: The system that improves by learning from data. b) Key difference: Rule-based AI uses predefined rules to make decisions, whereas learning-based AI learns patterns from data and improves over time.	
13	Reason for split: To check how well the model performs on unseen data. Training data is used to teach the model; testing data is used to evaluate its performance. Suitable ratio: Common ratios: 80% training, 20% testing (or 70/30). This ensures the model learns well but still has enough unseen data for testing.	
14	Possible Problem: The model may give incorrect or unreliable predictions because its accuracy was never verified. It might work well on the training data but fail on new, unseen data (overfitting). How a Test Set Helps: A test set helps to check how well the model performs on unseen data. It helps identify whether the model needs improvement before real-world use.	
15	Image recognition features are used, which are a subset of computer vision. The camera captures the image of the dog and preprocesses the image to enhance the image quality using noise reduction and correcting lighting. After that, the feature extraction helps to analyze the various features in the image, like edges, shapes, colors, and textures. This technique is known as edge detection. after that convolutional neural networks (CNNs) recognize patterns in the image and process through multiple layers.	
16	a) Lemmatization – for converting words to their meaningful root form (e.g., “running” → “run”) b) Stemming – for reducing words to their base forms roughly (e.g., “cats” → “cat”)	
	Answer any 3 out of given 5 question on employability skills(answer each question in50-80 words)	3X4=12

17	To develop an AI project, the AI Project Cycle provides us with an appropriate framework which can lead us towards the goal. The AI project cycle is the cyclical process followed to complete an AI project. Main Steps in the AI Project Cycle: 1. Problem Scoping: Clearly define the problem (e.g., predicting traffic congestion). 2. Data Acquisition : Gather relevant data such as traffic sensor data, GPS data, historical traffic records. 3. Data Exploration/validation: Clean, organize, and preprocess the collected data for analysis. 4. Modelling: Select and develop an AI/ML model suitable for prediction.& Train the AI model using the prepared dataset. 5. Model Evaluation: Test the model using unseen data to check its accuracy. 6. Deployment: Implement the AI system for real-time traffic prediction.	
18	Expected Answer: a) Neural Network: A neural network is an AI model inspired by the human brain, capable of learning patterns from data to make predictions or decisions. It is suitable for Ravi's task because it can analyze multiple inputs (study hours, attendance, past performance) and predict outcomes (pass/fail) accurately. b) Three layers and their roles: 1. **Input Layer**: Takes Ravi's data (study hours, attendance, past performance) as input for the network. 2.**Hidden Layer:** Processes the input data, identifies patterns and relationships between different factors affecting student performance.3.**Output Layer**: Provides the prediction result—whether a student is likely to pass or fail.	
19	i) It is Supervised Learning (Regression) since input features like size, rooms, and location predict a continuous target value—price. Here: Inputs (features) = Size, No. of Rooms, Location Output (label) = Price Supervised learning is a type of machine learning where the model is trained using a labeled dataset — that means each training example includes both: • Input (features) • Output (target label) The model learns the relationship between input and output so that it can predict the output for new, unseen data. **ii)Describe the other main categories of supervised machine learning models and how they differ from this one**.1. Regression • Purpose: Predict continuous numeric values • Output: A real number • Examples: o Predicting house prices o Estimating sales revenue o Predicting temperature Classification • Purpose: Predict categorical values (classes) • Output: A class label • Examples: o Email → "Spam" or "Not Spam" o Tumor → "Benign" or "Malignant" o Customer → "Will Buy" or "Won't Buy" 1 Mark for each	

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Confusion Matrix		Reality	
		Yes	No
Prediction	Yes	**150**	**50**
	No	**50**	**750**

Precision=(TP/(TP+FP))100
=150/(150+50)
=150/200
=0.75
Recall=TP/(TP+FN)
=150/(150+50)
=150/200
=0.75
F1 Score = 2 * Precision * Recall / (Precision + Recall)
=2 * 0.75 * 0.75 / (0.75+0.75)
=0.75
=75%
1 mark for confusion matrix
1/2mark for formula of each f1 score,precision, recall
1/2mark for putting values in each f1 score,precision, recall

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Answer: Text normalization helps to convert text into a standard format to improve NLP processing, it include lowercasing, removing extra spaces, correcting spellings, stemming and lemmatization.

Step 1: Input text given
- Document 1: Akash and Ajay are best friends.
- Document 2: Akash likes to play football but Ajay prefers to play online games.

Step 2: Convert to lowercase
- Document 1: akash and ajay are best friends.
- Document 2: akash likes to play football but ajay prefers to play online games.

Step 3: Remove Extra space
- Document 1: akash and ajay are best friends.
- Document 2: akash likes to play football but ajay prefers to play online games.

Step 4: Tokenization
- Document 1 Tokens: akash, and, ajay, are, best, friends
- Document 2 Tokens: akash, likes, to, play, football, but, ajay, prefers, to, play, online, games

Step 5: Lemmatization
- Document 1: akash and ajay are best friend.
- Document 2: akash like to play football but ajay prefer to play online game.